Senior Advanced Manufacturing Engineer, Laser Processing & Precision Joining – NMPH – Fridley, MN (Onsite)

Medtronic · Fridley, Minnesota, United States of America

Job ID (this archive)	HI-135
Employer	Medtronic
Location(s)	Fridley, Minnesota, United States of America
Career stage (archive tag)	Experienced
Location group	Other U.S.
Majors (archive tags)	Mechanical Engineering, Materials Science & Engineering
Pay listed	$103,200 – $154,800 per year
Employment type	Full time
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Employer requisition	R76856
Recruiting status when checked	Open when checked (Oct 11, 2026). Employer note: Open — employer posting available Saved PDF = full employer posting text, reformatted for print.
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Captured	2026-10-11
Course sheet refs	MS28

> Saved October 11, 2026. PDF text snapshot of the live company posting. Original wording retained; web layout and navigation omitted. Live listings change and close; check the source page before applying.

Job description

We anticipate the application window for this opening will close on - 17 Oct 2026

Careers that change lives start here. Medtronic is a global leader in healthcare technology with a Mission to alleviate pain, restore health, and extend life. Our 95,000 employees work across more than 150 countries to put patients first - developing innovative medical technologies that improve the lives of 72+ million patients each year. Your unique talents will help shape the future of healthcare while building a career grounded in purpose, growth, and impact.

A Day in the LifeAt Medtronic Neuroscience, we bring innovation, expertise, and purpose together to improve outcomes and transform lives for people living with neurological, spine, and chronic pain conditions.

Medtronic Neuromodulation and Pelvic Health (NMPH) deliver innovative, evidence-based therapies that help restore function, relieve chronic conditions, and improve quality of life. Through personalized, minimally invasive neuromodulation treatments, we empower patients to regain independence, confidence, and control, enabling better outcomes and helping them live life more fully.

This role is part of NMPH Advanced Manufacturing Engineering function, supporting the development of next-generation implantable medical devices. The team partners closely with Research & Development, Design Engineering, Materials Engineering, and manufacturing organizations to evaluate manufacturing feasibility, develop innovative process technologies, and transition scalable manufacturing solutions from concept through commercialization.

Based in Fridley, Minnesota, this is an onsite position supporting hands-on laboratory, prototyping, and manufacturing development activities. The role focuses on advanced manufacturing technologies including laser processing, precision joining, hermetic sealing, and assembly processes used in high-reliability implantable medical device applications. Travel is expected up to 25% to support manufacturing partners and technology transfer activities.

The Senior Advanced Manufacturing Engineer will lead the development and optimization of advanced manufacturing processes supporting early-stage product development, rapid prototyping, and manufacturing feasibility for implantable medical devices. This role combines hands-on technical problem solving with process development and cross-functional collaboration to translate product concepts into robust manufacturing solutions.

Primary Responsibilities

Develop and optimize manufacturing processes involving laser welding, laser cutting, laser marking, laser texturing, resistance welding, and other precision joining technologies.

Lead feasibility studies, rapid prototyping activities, and structured experimentation to evaluate manufacturability, process capability, and technical risk.

Design and execute Design of Experiments (DOE), process characterization studies, and test methods to establish process windows and correlate manufacturing parameters to product performance.

Develop manufacturing solutions for miniature components, implantable assemblies, hermetic packages, leads, and other high-reliability medical device applications.

Partner with Design, Materials, Technology, and Research & Development teams to influence design decisions, material selection, tolerancing strategies, and manufacturing methods.

Develop and troubleshoot hermetic sealing and leak-testing processes, including helium leak testing, to support implantable device performance requirements.

Create or support tooling, fixturing, precision alignment, component handling, and assembly solutions used for prototype and development builds.

Support process transfer, manufacturing scale-up, and technology implementation with internal manufacturing sites and external manufacturing partners.

Required Qualifications

Bachelor’s degree in Mechanical Engineering, Manufacturing Engineering, Materials Engineering, Welding Engineering, or a related engineering discipline and 4+ years of experience in manufacturing engineering, OR

Master’s degree in Engineering discipline and 2+ years of experience in manufacturing engineering, OR

PhD in Engineering discipline and 0+ years of experience in manufacturing engineering.

Hands-on experience with laser processing, resistance welding and joining technologies, precision manufacturing processes, or related advanced manufacturing methods.

Experience developing manufacturing processes through structured experimentation, process characterization, and technical problem solving.

Experience analyzing process performance and using data-driven approaches to establish process capability and manufacturing solutions.

Preferred Qualifications

Experience with laser welding, cutting, texturing, marking, ablation, micro-machining, or related laser-material processing technologies.

Experience with parallel-gap welding, step-gap welding, or other precision joining processes.

Experience supporting implantable medical devices, miniature assemblies, or other highly regulated, high-reliability products.

Experience developing and applying test methods to evaluate weld and joint quality, including mechanical testing, microscopy, cross-sectional analysis, dimensional inspection, or related characterization techniques.

Experience with hermetic sealing, helium leak testing, package integrity assessment, CAD/CAM systems, tooling and fixture design, new product introduction, and transfer to manufacturing.

For Baccalaureate degrees earned outside of the United States, a degree that satisfies the requirements of 8 C.F.R. § 214.2(h)(4)(iii)(A) is required.

Physical Job Requirements

The above statements are intended to describe the general nature and level of work being performed by employees assigned to this position, but they are not an exhaustive list of all the required responsibilities and skills of this position.

The physical demands described within the Responsibilities section of this job description are representative of those that must be met by an employee to successfully perform the essential functions of this job. Reasonable accommodations may be made to enable individuals with disabilities to perform the essential functions. For Office Roles: While performing the duties of this job,

the employee is regularly required to be independently mobile. The employee is also required to interact with a computer, and communicate with peers and co-workers. Contact your manager or local HR to understand the Work Conditions and Physical requirements that may be specific to each role.

U.S. Work Authorization & Sponsorship

At Medtronic, we are committed to fostering an environment where employees can thrive and make a meaningful impact. In alignment with our enterprise-wide workforce planning approach, U.S. work authorization sponsorship (H-1B, TN, J, etc.) is offered exclusively for Principal-level roles and above, where specialized expertise aligns with long-term business needs. Roles below the Principal level require candidates to possess unrestricted U.S. work authorization at the time of hire and for the duration of employment.

Recruitment Fraud Alert

We are aware of phishing scams targeting job seekers. Please keep the following in mind:

Apply only through official Medtronic channels. All legitimate Medtronic recruiting communications come from approved Medtronic platforms and official @medtronic.com email addresses.

Medtronic will never ask for payment or sensitive personal information (such as bank account or Social Security details) during early stages of the hiring process. Any such requests are not legitimate.

If you receive a suspicious message claiming to be from Medtronic, do not respond, click links, or open attachments.

If you have any questions, concerns regarding the authenticity of a communication alleged to have been made by or on behalf of Medtronic, please contact us immediately at AskHR@medtronic.com.

Benefits & Compensation

Medtronic offers a competitive Salary and flexible Benefits Package

A commitment to our employees lives at the core of our values. We recognize their contributions. They share in the success they help to create. We offer a wide range of benefits, resources, and competitive compensation plans designed to support you at every career and life stage.

Salary ranges for U.S (excl. PR) locations (USD):$103,200.00 - $154,800.00

This position is eligible for a short-term incentive called the Medtronic Incentive Plan (MIP).

The base salary range is applicable across the United States, excluding Puerto Rico and specific locations in California. The offered rate complies with federal and local regulations and may vary based on factors such as experience, certification/education, market conditions, and location. Compensation and benefits information pertains solely to candidates hired within the United States (local market compensation and benefits will apply for others).

The following benefits and additional compensation are available to those regular employees who work 20+ hours per week: Health, Dental and vision insurance, Health Savings Account, Healthcare Flexible Spending Account, Life insurance, Long-term disability leave, Dependent daycare spending account, Tuition assistance/reimbursement, and Simple Steps (global well-being program).

The following benefits and additional compensation are available to all regular employees: Incentive plans, 401(k) plan plus employer contribution and match, Short-term disability, Paid time off, Paid holidays, Employee Stock Purchase Plan, Employee Assistance Program, Non-qualified Retirement Plan Supplement (subject to IRS earning minimums), and Capital Accumulation Plan (available to Vice Presidents and above, or subject to IRS earning minimums).

Regular employees are those who are not temporary, such as interns. Temporary employees are eligible for paid sick time, as required under applicable state law, and the Employee Stock Purchase Plan. Please note some of the above benefits may not apply to workers in Puerto Rico.

Further details are available at the link below:

Medtronic benefits and compensation plans

It is the policy of Medtronic to provide equal employment opportunity (EEO) to all persons regardless of age, color, national origin, citizenship status, physical or mental disability, race, religion, creed, gender, sex, sexual orientation, gender identity and/or expression, genetic information, marital status, status with regard to public assistance, veteran status, or any other characteristic protected by federal, state or local law. In addition, Medtronic will provide reasonable accommodations for qualified individuals with disabilities.

If you are applying to perform work for Medtronic, Inc. (“Medtronic”) in any position which will involve performing at least two (2) hours of work on average each week within the unincorporated areas of Los Angeles County, you can find here a list of all material job duties of the specific job position which Medtronic reasonably believes that criminal history may have a direct, adverse and negative relationship potentially resulting in the withdrawal of a conditional offer of employment. Medtronic will

consider for employment qualified job applicants with arrest or conviction records in accordance with the Los Angeles County Fair Chance Ordinance for Employers and the California Fair Chance Act.

Notes for ECS 1100 students

Course spreadsheet notes

Skills mentioned: Laser processing; precision joining; hermetic sealing; process development.

Materials Science

Materials Engineering explicitly listed; manufacturing of implantable devices.

Degree: Bachelor’s + 4 years OR master’s + 2 years OR PhD + 0 years

Experience / eligibility: Relevant manufacturing engineering experience per degree route.

Other majors named: ME, Manufacturing, Materials, Welding Engineering or related

Career stage: Experienced / senior / Senior professional / PhD route

Salary evidence: Salary ranges for U.S (excl. PR) locations (USD):$103,200.00 - $154,800.00